

Statistics Notes: Diagnostic tests 1: sensitivity and specificity

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Abstract & Summary:

In many clinical and diagnostic situations, an evaluation or screening test is used to classify subjects into one of two categories, such as positive or negative for a condition or defect. When a new diagnostic test or classification procedure is evaluated, its performance is compared with an authentic reference standard (often termed the 'gold standard').

Definitions and Contingency Formulation:

The diagnostic classification results can be represented in a standard two-by-two contingency table comparing the test predictions against the authentic reference status:

	Reference Standard (Condition Present)	Reference Standard (Condition Absent)
Test Positive:	True Positive (a)	False Positive (b)
Test Negative:	False Negative (c)	True Negative (d)

1. Sensitivity:

Sensitivity is the proportion of true positives correctly identified by the test among all subjects who actually possess the condition:

$$\text{Sensitivity} = a / (a + c) = \text{True Positives} / (\text{True Positives} + \text{False Negatives})$$

2. Specificity:

Specificity is the proportion of true negatives correctly identified by the test among all subjects who do not possess the condition:

$$\text{Specificity} = d / (b + d) = \text{True Negatives} / (\text{True Negatives} + \text{False Positives})$$

Interpretation and Properties:

Both sensitivity and specificity are fundamental diagnostic proportions ranging from 0 to 1 (or 0% to 100%). Sensitivity quantifies the diagnostic test's ability to avoid false negatives (detecting actual cases), while specificity quantifies its resistance to false alarms (avoiding false positives on clean or compliant cases).

Standard confidence intervals can be computed for both proportions using the standard error formula for binomial proportions. Sensitivity and specificity reflect intrinsic discriminative properties of the test procedure across the evaluated sample.